

5. $x^2 y^3 - 2y^2 - 3x + 7 = 0$; $\frac{dy}{dx} = \dots$

$$\frac{dy}{dx} [x^2 y^3 - 2y^2 - 3x + 7] = \frac{dy}{dx} 0$$

$$= \frac{dy}{dx} x^2 y^3 - \frac{dy}{dx} 2y^2 - \frac{dy}{dx} 3x + \frac{dy}{dx} 7 = 0$$

$$= 2xy^3 + 3x^2 y^2 \frac{dy}{dx} - 4y \frac{dy}{dx} - 3 + 0 = 0$$

$$= (3x^2 y^2 - 4y) \frac{dy}{dx} = 3 - 2xy^3$$

$$\frac{dy}{dx} = - \frac{(2xy^3 - 3)}{3x^2 y^2 - 4y}$$

6. $2x = \frac{x-y}{x+y}$ $\frac{dy}{dx} = \dots$

$$\frac{dy}{dx} 2x = \frac{dy}{dx} \left(\frac{x-y}{x+y} \right)$$

$$2 = \frac{\left[\frac{dy}{dx} (x-y) \right] (x+y) - (x-y) \left[\frac{dy}{dx} (x+y) \right]}{(x+y)^2}$$

$$2(x+y)^2 \left[1 - \frac{dy}{dx} \right] (x+y) - (x-y) \left[1 + \frac{dy}{dx} \right]$$

$$2(x+y)^2 = x+y - x \frac{dy}{dx} - y \frac{dy}{dx} - \left[x-y + x \frac{dy}{dx} - y \frac{dy}{dx} \right]$$

$$2(x+y)^2 = x-x+y+y - x \frac{dy}{dx} - x \frac{dy}{dx} - y \frac{dy}{dx} + y \frac{dy}{dx}$$

$$2(x^2 + 2xy + y^2) = 2y - 2x \frac{dy}{dx}$$

$$2x^2 + 4xy + 2y^2 - 2y = -2x \frac{dy}{dx}$$

$$\frac{dy}{dx} = -x - 2y - \frac{y^2}{x} + \frac{y}{x}$$

Jadi, $\frac{dy}{dx} = -x - 2y - \frac{y^2}{x} + \frac{y}{x}$

g. Diberikan kurva $C \rightarrow ax^2 + y^2 = b$ melalui $(1,1)$.

Garis singgung di $(1,1) \rightarrow 2x + y = 3$

• menentukan gradien garis singgung :

$$2x + y = 3$$

$$y = -2x + 3$$

$$m_s = -2$$

• m_s adalah $\frac{dy}{dx}$ di $x=1$; $y=1$

$$\frac{dy}{dx} [ax^2 + y^2] = \frac{dy}{dx} b$$

$$2ax + 2y \frac{dy}{dx} = 0$$

$$2y \cdot y' = -2ax$$

$$y' = \frac{-ax}{y^2}$$

$$y' = \frac{-ax}{y^2}$$

Substitusi $(1,1) \rightarrow y' = \frac{-a}{1^2}$

• Karena $m_s = y'$; maka :

$$m_s = y'$$

$$-2 = \frac{-a}{1^2}$$

$$a = 2$$

• Substitusi a ke $ax^2 + y^2 = b$

$$2x^2 + y^2 = b$$

Substitusi $(1,1)$ ke persamaan

$$2(1)^2 + (1)^2 = b$$

$$b = 2 + 1$$

$$b = 3$$

Jadi, nilai a adalah 2 dan b adalah 3

13). Dapatkan $f'(x)$ dari persamaan :

a. $f(x) = (x^3 + 2x^2 - x)^7$

$$f'(x) = 7(x^3 + 2x^2 - x)^6 (3x^2 + 4x - 1)$$

$$= 7(3x^2 + 4x - 1)(x^3 + 2x^2 - x)^6$$

Jadi, $f'(x) = 7(3x^2 + 4x - 1)(x^3 + 2x^2 - x)^6$

b. $f(x) = (2x^4 - 3x^2 + x)^{33}$

$$f'(x) = 33(2x^4 - 3x^2 + x)^{32} (8x^3 - 6x + 1)$$

$$= 33(8x^3 - 6x + 1)(2x^4 - 3x^2 + x)^{32}$$

Jadi, $f'(x) = 33(8x^3 - 6x + 1)(2x^4 - 3x^2 + x)^{32}$

c. $f(x) = \left(\frac{2}{x^2} - x^5 \right)^{-3}$

$$\rightarrow f(x) = (2x^{-2} - x^5)^{-3}$$

$$f'(x) = -3(2x^{-2} - x^5)^{-4} (-4x^{-3} - 5x^4)$$

$$= \frac{-3}{\left(\frac{2}{x^2} - x^5 \right)^4} \left[\frac{-4}{x^3} - 5x^4 \right]$$

Jadi, $f'(x) = \frac{-3}{\left(\frac{2}{x^2} - x^5 \right)^4} \left[\frac{-4}{x^3} - 5x^4 \right]$

$$d). f(x) = \frac{1}{(x^5 - x^2 + x)^9}$$

$$f(x) = (x^5 - x^2 + x)^{-9}$$

$$f'(x) = -9(x^5 - x^2 + x)^{-10} (5x^4 - 2x + 1)$$

$$= \frac{-9(5x^4 - 2x + 1)}{(x^5 - x^2 + x)^{10}}$$

$$14. a. y = \left(\frac{x-3}{2x+1} \right)^3$$

$$y' = 3 \left(\frac{x-3}{2x+1} \right)^2 \left[\frac{\left(\frac{dy}{dx}(x-3) \right)(2x+1) - (x-3) \left(\frac{dy}{dx}(2x+1) \right)}{(2x+1)^2} \right]$$

$$y' = 3 \left(\frac{x-3}{2x+1} \right)^2 \left[\frac{(2x+1) - 2(x-3)}{(2x+1)^2} \right]$$

$$y' = \frac{3(x-3)^2 [2x+1-2x+6]}{(2x+1)^4}$$

$$= \frac{3(x^2 - 6x + 9) [7]}{(2x+1)^4}$$

$$= \frac{21x^2 - 126x + 189}{(2x+1)^4}$$

Jadi $\frac{dy}{dx}$ dari $y = \left(\frac{x-3}{2x+1} \right)^3$ adalah

$$y' = \frac{21x^2 - 126x + 189}{(2x+1)^4}$$

$$b. y = \frac{(2x-3)^2}{(3x^2+1)^7}$$

$$y = \frac{(4x^2 - 12x + 9)}{(3x^2+1)^7}$$

$$y' = \frac{\left(\frac{dy}{dx} 4x^2 - 12x + 9 \right) (3x^2+1)^7 - (4x^2 - 12x + 9) \left(\frac{dy}{dx} (3x^2+1)^7 \right)}{[(3x^2+1)^7]^2}$$

$$= \frac{(8x - 12) (3x^2+1)^7 - (4x^2 - 12x + 9) 7(3x^2+1)^6 \cdot 6x}{(3x^2+1)^{14}}$$

$$= \frac{[3x^2+1]^6 ((8x-12)(3x^2+1) - (4x^2-12x+9) \cdot 42x)}{(3x^2+1)^{14}}$$

$$= \frac{(24x^3 - 36x^2 + 8x - 12) - (168x^3 - 504x^2 + 378x)}{(3x^2+1)^8}$$

$$= \frac{-144x^3 + 468x^2 - 370x}{(3x^2+1)^8}$$

Jadi $\frac{dy}{dx} = \frac{-144x^3 + 468x^2 - 370x}{(3x^2+1)^8}$

$$15). f'(x) = \sqrt{3x+4} \quad g(x) = x^2 - 1$$

$$F(x) = f(g(x))$$

$$F'(x) = [f'(g(x))] \cdot g'(x)$$

$$= (\sqrt{3g(x)+4}) \cdot 2x$$

$$= 2x \sqrt{3x^2 - 3 + 4}$$

$$= 2x \sqrt{3x^2 + 1}$$

Jadi, $F'(x) = 2x \sqrt{3x^2 + 1}$